Anmerkung Conditions Report

Rule engine covering Dachser, K+N, DHL Express, Wackler & Honold · source: assets/anmerkung.js · generated 2026-10-16

How a row is classified

Each worksheet row is read column-wise into the five *process** functions. A row is only classifiable when its **Stat_Freigabe** cell equals **10** (released); otherwise the row is skipped (Wackler still runs its Kontierung check). Each forwarder compares its forwarder-specific **Differenz** columns (FR, MT, SNK, EXP, TZ, …) against a **firing threshold**:

```
hasErr(v, t) = |v| > t
```

A delta that clears the threshold is called a "trigger" and contributes a label. All labels a row gathers are joined with **" // "** into the **Anmerkung** cell (deduplicated by join). Some rules *return early* or block later rules, because a single finding (e.g. a duplicate Fremdnummer booking) makes every other delta on the row spurious.

Thresholds

Constant	Value
Firing threshold — Dachser	|diff| > ? (default 0.08)
Firing threshold — K+N	|diff| > ? (default 0.09)
Firing threshold — DHL Express	|diff| > ? (default 0.04)
Firing threshold — Wackler	|diff| > ? (default 0.09)
Persisted per-user	localStorage key <font face='Courier' size='8'>anme
Kontierung? checks	disabled globally — KONTIERUNG_ENABLED = fal

1. Dachser — processDachser

Gate: Stat = 10. Reads C502/C503 (Einlagern/Auslagern), SNK, ZZ, SAM, DGR, EXP, MT, TZ, C38L, FR with weight/tier logic on the DACHSER_BP rate-card tiers (50 kg steps → 10 000 kg, open 999 999 bucket). Cascade order matters — the first rule that fires colours the rest.

1.0 Blank-Tarif accounting rows (early return)

Condition	Label written to Anmerkung
TARIF cell empty AND |FR| > T; no SACH, no SERV_ART, no other diff (MT/TZ/EXP/SNK/ZZ/DGR/LG/AV/SAM/SBFU/C38L)	`VORHOLUNG`
same empty-TARIF but SACH + SERV_ART are set	`Fremdnummer <row's ReferenzNr> bereits berechnet in RE00123xxx, ok? (falls back to the 5034xxx placeholder when ReferenzNr is empty; the Beleg it was charged in stays a placeholder, it names a document outside this workbook)`
empty TARIF + FR trigger + departure country non-DE	`kein Tarif für <Land>`
empty TARIF + FR trigger + domestic lane above 10 t (Brutto kg > 10000)	`Kein Tarif für DE >10 to, ok?`

1.1 Direct-cost labels (each fires independently)

Condition	Label written to Anmerkung
C196 Differenz over threshold (Kalkulationskosten)	`Kalk. Kosten Zustellung (emitted as the leading segment)`
C502 Kosten DL has a non-zero value	`Einlagern`
C503 Kosten DL has a non-zero value	`Auslagern`
LG Differenz over threshold	`Lagergeld`
AV Differenz over threshold	`Gebühr für vergeblichen Abholversuch`
ZZ Differenz over threshold	`2. Zustellung`
SAM Differenz over threshold	`Samstagzustellung`
DGR Differenz over threshold (skipped when TARIF is zero)	`Gefahrgut-Zuschlag`
EXP Differenz over threshold	`Produktzuschlag, or Terminzuschlag when EXP Kosten DL = 95`
ZABF Differenz over threshold	`Einfuhrzollabfertigung`
SBFU Differenz over threshold	`SBfU-Bescheinigung f. Umsatzsteuerzwecke`
C38L Differenz over threshold	`Maut von/bis NL`

1.2 SNK — switch on billed SNK_DL

Condition	Label written to Anmerkung
SNK_DL = 190 or 95	`AUSFALLFRACHT`
SNK_DL = 130 or 390	`Standgeld`
SNK_DL = 75 + SNK_TARIF > 0 + SNK diff over T	`Differenz Hebebuehnen-Zuschlag`
SNK_DL = 75 + SERV_ART = K1AV	`Speditionskosten gem. Text`
SNK_DL = 75 (no tariff backing)	`Ausfallfracht/Schadensersatz`
SNK_DL = 80 + SERV_ART = K1AV	`Speditionskosten gem. Text // Admin Zeitfensterbuchung Handel`

SNK_DL = 11 (+ diff over T)	`Differenz Telefonische Zustellankündigung - Laderaumzuschlag`
SNK_DL = 14 (+ diff over T, TARIF not zero)	`Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag (K1AV → Differenz Laderaumkostenentwicklung)`
SNK_DL = 5 + SERV_ART = K1AV (SNK_TARIF=0)	`Admin Zeitfensterbuchung Handel`
SNK_DL = 5 + SERV_ART = K1AV (SNK_TARIF>0)	`Differenz Admin Zeitfensterbuchung Handel - Laderaumzuschlag`
SNK_DL = 5 + diff over T + domestic lane (Abg.-Land blank or DE); skipped when TARIF zero	`Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag`
SNK_DL = 5 + diff over T + import lane (Abg.-Land not DE, Empf.-Land DE)	`Differenz Laderaumkostenentwicklung`
SNK_DL = 9 (+ diff over T, TARIF not zero)	`Differenz Telefonische Zustellterminvereinbarung - Laderaumzuschlag`
other SNK_DL + SERV_ART = K1AV, or non-integer SNK_DL whose SNK_DIFF > T (a residual equal to the 0.08 threshold does not qualify)	`Differenz Laderaumkostenentwicklung`
K1AV and SNK_DL is a whole 65-multiple (≥ 65) and not one of the Standgeld values (130, 390)	`Wartezeit <n>h á 65 EUR, ok? (the headline alone, no admin-fee suffix)`
K1AV and SNK_DL − 5 ≥ 65, residual a whole 65-multiple	`Wartezeit <n>h á 65 EUR, ok? // Differenz Admin Zeitfensterbuchung Handel - Laderaumzuschlag (the bundled admin fee is still owed, so the phrase is not emitted alone)`
K1AV and SNK_DL − 5 ≥ 65, residual cents-bearing and not a 65-multiple	`<amount> Kosten für 2.Zustellung etc. ok?`
other SNK_DL (+ diff over T)	`Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag`
Non-integer SNK_DL and SNK_DIFF rounds to 5/9/11/14 (±0.05)	`code re-derived from SNK_DIFF, then the switch above`

1.3 EXP + others

Condition	Label written to Anmerkung
EXP diff over T and EXP_DL = 95	`Terminzuschlag`
EXP diff over T (otherwise)	`Produktzuschlag`
Empf.-Ort = LONDON, PLZ leading letter	`Zone korrekt berechnet? (+ Einfuhrzollabfertigung when EXP clean)`
ZABF Differenz over threshold (Import-Zollabfertigung line)	`Einfuhrzollabfertigung`

1.4 FR — weight/tier decision tree (after SNK)

Condition	Label written to Anmerkung
SERV_ART = K1AU and the rate cards return Sonderfahrt for the lane (FR Kosten DL − 180/150/120/90 lands on a NEW-card cell)	`Sonderfahrt`
Anz. Sdg > 1 (multi-consignment)	`hätte gebündelt werden können?`
REFERENZ3 = ZW (deviating intermediate consignee)	`Differenz aufgrund abweichender Zwischenempfänger <PLZ Ort>`

SERV_ART = K1AS	`Sonderfahrt`
SERV_ART = K1AU and |FR − 90| ≤ 0.1	`"Sonderfahrt" 90 EUR doppelt berechnet?`
weights equal or same tier; FR > 1 on an international lane (non-DE origin or destination)	`Sonderfahrt, or Dachser berechnet die Kosten nach dem bisherigen Tarif — decided by the rate cards: (FR Kosten DL − S) landing on a NEW-card cell for S in 180/150/120/90 gives Sonderfahrt, otherwise FR Kosten lt. Tarif landing on an OLD-card cell (timed by 1.15 on imports) gives the bisherigen wording. Ratecards are encrypted (assets/dachser-ratecards.enc.json); without the passphrase the bisherigen wording stands.`
weights equal or same tier; FR > 1 (or ≥ 0.05 with no other label); ZZ diff = 35	`Fracht Differenz (ZZ=35)`
weights equal or same tier; FR > 1 (or ≥ 0.05 with no other label); otherwise	`Frachtdifferenz`
weights equal or same tier; FR < –1.0	`Differenz aufgrund abweichender Gewichte`
both weights known, different tier, FR 0–1	`Frachtdifferenz`
both weights known, different tier (FR ≥ 1 or negative)	`Differenz aufgrund abweichender Gewichte`
no weight basis, FR negative (or < 1.0)	`Differenz Frachtzu/ abschlag`
no weight basis, FR positive	`Differenz aufgrund von abweichendem Gewicht`
Condition	**Label written to Anmerkung**
SNK_DL = 14 + SNK diff over T (footer)	`Abholterminvereinbarung`
row has no other label, only TZ over T	`Differenz treibstof`

2. K+N — processKN

Gate: Stat = 10. Tiers from KN_BP (like Dachser but collapses the 8500/9500 steps and opens at 99 999). FR (freight) is the lead column; EXP/MT/SNK/TZ add their own labels.

Condition	Label written to Anmerkung
FR over T and ReferenzNr lists multiple docs (contains ',') — early return	`hätte gebündelt werden müssen, ok?`
FR over T, TARIF blank or '-' — flat-rate freight	`Pauschalfracht`
FR over T, Amazon recipient, VKG and VKG_DL in the same weight tier	`hätte nach Amazon Tarif abrechnen müssen`
FR over T, Amazon recipient, VKG and VKG_DL cross tiers	`Differenz aufgrund abweichender Gewichte`
FR > 0 and SNK_DL = 5, same tier	`hätte nach Amazon Tarif abrechnen dürfen`
FR > 0 and SNK_DL = 5, cross tier	`Differenz aufgrund abweichender Gewichte`
any other FR over T	`Differenz aufgrund abweichender Gewichte`
EXP over T	`FIXTERMIN`
MT over T	`Mautdifferenz`

SNK cascade — exact matches first, then tolerance fallback

Condition	Label written to Anmerkung
SNK diff ≈ +9	`Avis, ok?`
SNK diff ≈ –9	`Differenz avis`
|SNK diff| ≈ 25	`Portalavisierung, ok?`
SNK diff over T and SNK_DL = 5 or 25	`Portalavisierung, ok?`
SNK diff over T and SNK_DL = 9	`Avis, ok?`
SNK diff over T and SNK_DL = 12	`B2C-zuschlag, ok?`
SNK diff over T and SNK_DL = 18 (unless Avis already)	`Avis, ok?`
SNK diff over T and SNK_DL = 34	`Portalavisierung, ok? // Avis, ok?`
SNK diff over T, code not matched	`SNK Differenz`
TZ over T AND neither FR nor MT fired	`Differenz treibstoff`
KOST or SACH blank/'–' (disabled)	`Kontierung?`

3. DHL Express — processDHL

Gate: Stat = 10. Uses a **blocking chain**: the first group (EL/FR/PAL/OW/YO/YL/ND/SF + SNK) sets block, and only when nothing blocked does the AC/MT/NX/OS group fire.

Condition	Label written to Anmerkung
TARIF present but value 0 (raw contains '0' or is '-')	`Fremdnummer xxx bereits berechnet in RExxx, ok?.`
EL Differenz over T	`EXTENDED LIABILITY, ok?`
FR Differenz over T	`Differenz aufgrund von abweichendem Gewicht/Volumen`
PAL Differenz over T	`nicht stapelbar ok?`
OW Differenz over T	`Differenz overweight, ok?`
YO positive and divisible by 15	`Non conveyable piece-weight ok?`
YO over T (not a 15-multiple)	`non conveyable piece ok?`
YL over T	`Non-conveyable piece irregular ok?`
ND over T	`Neutral delivery ok?`
SF over T	`Direct signature ok?`
SNK = 5	`DIRECT SIGNATURE , ok?`
SNK = 25	`Limited quantities ok?`
SNK = 30	`Elevated Risk, ok?`
SNK = 60	`Eelevated risk ok? // Restricted destination ok?`
SNK over T, code not matched	`SNK Differenz`
below-block group — AC = 11	`Addres Correction, ok?`
below-block group — AC over T	`Address Correction ok?`
below-block group — MT over T	`Mautdifferenz`
below-block group — NX over T	`demand surcharge ok?`
below-block group — OS over T	`Oversize piece ok?`
no other label, TZ over T only	`Differenz treibstof`

4. Wackler — processWackler

Gate: Stat = 10 (otherwise only the Kontierung check runs). Fourteen numbered rules evaluated in order with several early returns. Weight-tier logic uses the Wackler rate-card breakpoints (50 kg steps → 10 000 kg, open 999 999 bucket) shared by the national and international rate sheets.

Terminal (early-return) rules

Condition	Label written to Anmerkung
TARIF '-' or '0' (non-empty), or blank + FR/MT/TZ signal — duplicate booking; all delta rules suppressed	`Fremdnummer xxx bereits berechnet in RExxx, ok?`
TARIF blank + SNK signal + no FR/MT/TZ + no known SNK/AVIS code — storage fee	`Lagergeld, ok?`
SNK >= 1.0 × TARIF + no FR/MT/TZ + no codes (placeholder tariff)	`Pauschalfracht, ok?`

AVIS surcharge codes

Condition	Label written to Anmerkung
AVIS = 7.5 / 8.5 / 6.5 / 8.7 (sign-insensitive)	`Avis, ok?`
AVIS = –8.7 (should have used the cheap telephonic rate)	`hätte Avisgebühr telefonisch abrechnen dürfen`
AVIS = 1 (per-shipment Avisnachweis line)	`Differenz avis, ok?`

SNK surcharge code book (sign-insensitive, tolerance ±0.5)

Condition	Label written to Anmerkung
SNK ≈ 38	`NL-FIX`
SNK ≈ 11.5 (±0.1)	`hätte B2C-Line abrechnen dürfen`
SNK ≈ 22	`2. Zustellung ok?`
SNK ≈ 25	`Terminzustellung`
SNK ≈ 80 on a weighed shipment (weight-gated)	`Terminzustellung`
SNK ≈ 170	`Terminzustellung`
SNK ≈ 180	`Terminzustellung, ok?`
SNK ≈ 43	`2.Zustellung ok?`
SNK ≈ 289	`Umverfügung`
SNK ≈ 49	`NL-12, ok?`
SNK ≈ 113	`NL-SPEZ, ok?`
Empf. = 88499 RIEDLINGEN	`Return, ok?`
SNK ≈ 16	`NIGHTFIX`
SNK ≈ 81	`NL-10, ok?`
SNK ≈ 250	`"Zusatzkosten 250 €", ok?`

Gewichte / bundling decision tree (FR + both weights real)

Condition	Label written to Anmerkung
multi-ref, weights 1–5% apart (same or cross tier)	`hätte gebündelt werden müssen`
VKG and VKG_DL in different rate tiers	`Differenz aufgrund abweichender Gewichte`
same tier, ≥ 6 references	`hätte gebündelt werden müssen`
pallet volume (colli×285) ≥ 10 000 kg or ≥1.05×VKG, tiers apart	`Differenz aufgrund abweichender Gewichte`

same tier, ≥ 3 references (fallback)	`Differenz aufgrund abweichender Gewichte`
otherwise (same-tier rate-card case)	`Wackler rechnet Frachtrate für <tier>kg ab`
cross-tier, multi-ref, VKG_DL ≤ 20% of VKG — TERMINAL	`hätte gebündelt werden müssen (alone)`
cross-tier, multi-ref, FR > 0, weights within 20%	`hätte gebündelt werden müssen`
cross-tier, far weights or FR credit	`Differenz aufgrund abweichender Gewichte`
same-tier multi-ref with unequal whole-kg weights	`hätte gebündelt werden müssen`
same-tier multi-ref with (near-)identical weights, non-integer (volumetric) VKG	`Wackler rechnet Frachtrate für <floor tier>kg ab`
no real weights + multi-ref + FR	`hätte gebündelt werden müssen`

Fallback / additive rules

Condition	Label written to Anmerkung
FR over T, no Gewichte/bundling/return fired	`Frachtdifferenz`
MT over T	`Mautdifferenz`
|SNK| ≥ 5.0, unrecognised code, not bundled	`SNK Differenz`
SNK ≈ –150 (±2)	`hätte Hebebühne abrechnen dürfen`
|TZ| ≥ 2.0, not on same-tier 'Wackler rechnet' row, not an FR credit — on a Hebebühne credit row the fuel gate queries the surcharge instead of flagging it	`Differenz Energiezuschlag // Dieselzuschlag ok?`
NL-FIX + FR + blank KOST/SACH — 'Zone korrekt?' prepended, Frachtdifferenz stripped	`Zone korrekt? // …`
only rule left is TZ over T	`Differenz Energiezuschlag`
KOST & SACH both blank/X (disabled)	`Kontierung?`

Wackler constants

Constant	Value
|TZ| trigger (rule 11)	2.0
|SNK| noise floor (rule 10)	5.0
SNK ≈ –150 ± 2 → Hebebühne	150 ± 2.0
SNK ≥ ratio × TARIF → Pauschalfracht	× 1.0
SNK ≈ 80 ± 0.5 → Terminzustellung (weighed)	80 ± 0.5
Multi-ref same-tier upper bundle weight	1000
Cross-tier near band (bundling)	0.2
VKG_DL ≤ fraction of VKG → terminal bundling	0.2
Pallet volume estimate per collium (kg)	285
'Near-equal' weights band	0.01
Close-but-unequal bundle band	0.05
Same-tier ≥ N references → bundling	6

5. Honold — processHonold

Gate: Stat = 10, plus the OLD tariff must be unlocked. The tariff matrix is business data and ships encrypted (assets/honold-tariff.enc.json); until it is decrypted the engine *skips* Honold rows (null) rather than blanking an existing note it cannot re-derive. One rule: Volumen kg and Volumen kg DL in the same weight bracket mean the previous tariff is still being billed, a different DL bracket is a weight difference.

Condition	Label written to Anmerkung
Volumen kg and Volumen kg DL sit in the same Bis bracket of the OLD tariff (business test, no rates read), or Volumen kg DL is blank / 0	`Honold berechnet die Kosten nach dem bisherigen Tarif`
the two weights sit in different brackets	`Differenz aufgrund abweichender Gewichte`
blank / 0 FR Kosten DL, or a missing / 0 Volumen kg	`Differenz aufgrund abweichender Gewichte`

Honold constants

Constant	Value
Weight brackets	Bis column of the OLD tariff (50 … 10 000, last row c

How the output is consumed

The **Anmerkung** cell is a ' // '-joined list of independent rule outputs. The diff tooling (classifyDiff) compares the engine output against the auditor's ground-truth column after normalising phrase spelling/case/whitespace (normPhrase), and buckets each row as **correct / missed / overfired / wrong**. Phrase-level decomposition then reports which specific phrase the engine missed (*missing_phrases*) and which it over-fired (*extra_phrases*). That comes right from the A/B training pairs — a rule change is only accepted when it re-validates against the git-ignored Soll-Ist workbooks.

Control surface: scripts/anmerkung/verify.mjs smoke-tests that the five process* functions and window.AnmerkungV5 exist; the same periods feed the diagnostics facade. Firing thresholds and the Wackler constants above are read live out of assets/anmerkung.js at build time, so this report tracks the engine — but the cascade descriptions are maintained by hand in scripts/anmerkung/report_conditions.py and should be kept in sync when rules change.